

April Norton

I'm a fourth-year mechanical engineering and design student at Northeastern. I've co-oped at **Conair / Cuisinart** in product development and test engineering, and at the **Boston Water and Sewer Commission** in design and field survey. Robotics is where I'm headed next.

What I actually like is being hands on. Modeling a part and printing it overnight, taping thermocouples inside a prototype to find out what it really does, or cutting and sewing something until it fits. Same instinct, different material.



CURRENTLY

B.S. Engineering, Northeastern University
Class of 2028

CO-OPS

Conair / Cuisinart • Jan–Jun 2026

Boston Water & Sewer • Jan–Jun 2025

LOOKING TOWARD

Robotics

REACH ME

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SELECTED WORK • CONAIR / CUISINART

Six months, five problems, one very patient 3-D printer.

My co-op sat between design and test. I'd model a part in the morning, print it overnight, and find out at lunch the next day whether it survived contact with a potato. These five projects are the ones with the most to show: a consumer-safety redesign, a full-scale appearance mockup, a two-part silicone mold, an instrumented thermal study, and a competitive cook-quality benchmark.

Keeping fingers off a mandoline blade

A safety redesign of an existing mandoline slicer: iterating a hand guard and carriage in SolidWorks, printing each version, and testing it on real food until the blade stopped being the most interesting part of the tool.

CONSUMER SAFETY

SOLIDWORKS

FDM 3-D PRINTING

DESIGN ITERATION

FOOD TESTING

THE PROBLEM

A mandoline is a fixed razor blade with a ramp pointed at your hand. The stock guard on the unit we started with was easy to skip, because it didn't hold the food well, so the fastest way to use the tool was the way that put fingertips closest to the blade.

The design goal was blunt: make **the safe way also the easy way**. If the guard grips the food better than a bare hand does, nobody takes it off.

WHAT I DID

- Modeled a captive **food holder and carriage** in SolidWorks that rides on rails over the blade deck, so the food is driven forward without a hand ever crossing the cutting plane.
- Printed and tested successive versions: open collars first, then a fully enclosed dome with a spring-loaded, spiked pressure plate to bite into the food and take up slack as it shortens.
- Added a screw-mounted rail bracket so the carriage stays captured on the deck instead of lifting off mid-stroke.
- Tested each revision on real produce, checking grip on the last inch of a potato, the part of the stroke where injuries actually happen.

4+

Guard geometries printed and tested

0_{mm}

Hand travel across the blade plane, by design

2

Subsystems added: carriage rail + sprung pressure plate



Parts, mid-iteration. Printed collars and domes beside the stock hardware they had to mate with.



First rendition. Painter's tape as a false hinge. It sat inside the rails, where the julienne blades pop up and block it.



Second rendition. Arms on the rail, hinge in the existing track, and a much smaller gap to the deck, which is food you would otherwise waste.



The resolved guard. Thumb knob on top; sprung spiked plate underneath that follows the food down as it shortens.



Mounted on the carriage. The spring load meant the old track no longer held the pusher down, so these rails are new.



Food test. Judged on the last inch, where injuries happen.



In the hand. The grip puts the palm above and behind the blade.



Plan view. Clearance checked against the julienne teeth.



Full assembly. Installed on the production body for stroke testing.

A toaster oven you could hold before it existed

Translating a SolidWorks twin-cook oven model into a full-scale cardboard appearance mockup, accurate to a quarter inch, with a removable window fixture so the glass proportions could be swapped and argued about in person.

RAPID MOCKUP

SOLIDWORKS

HAND FABRICATION

DESIGN REVIEW

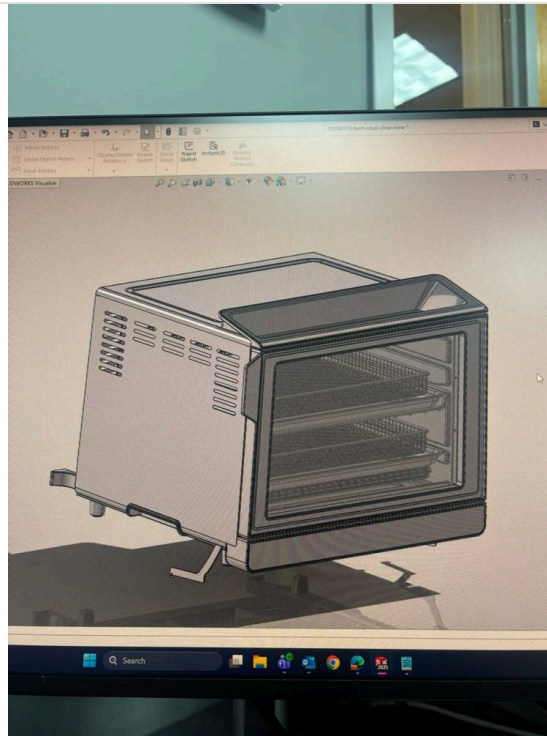
WHY CARDBOARD

A CAD render will tell you the oven is 16 inches wide. It will not tell you whether it fits under a cabinet, whether the door swing hits the backsplash, or whether the whole thing simply feels too big on a counter.

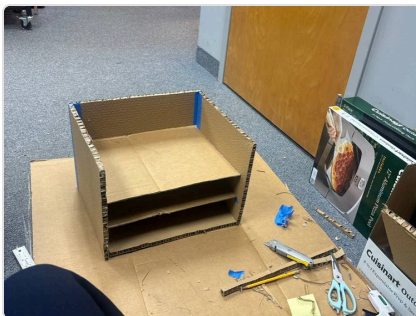
Cardboard answers those in an afternoon and costs nothing, so the expensive questions get settled before anyone commits to tooling.

HOW IT WAS BUILT

- Pulled dimensions straight off the SolidWorks twin-cook model and flattened them into cut patterns.
- Built the shell, cavity walls and both rack levels in double-wall board, holding $\pm\frac{1}{4}$ in against the CAD.
- Made the **front window aperture a removable fixture** so different glass sizes and bezel widths could be dropped in and compared side by side.
- Mocked the internal rack spacing so cookware clearance could be checked with real pans, not assumed.



The source model: the twin-cook oven assembly the mockup was measured from.



Build in progress. Shell and shelves, cut from CAD-derived patterns.



Cavity and racks. Real pan clearance, checked with real pans.



Finished. The front window lifts out so glass proportions can be swapped.

Designing the mold, not just the part

A two-plate injection mold designed from scratch in SolidWorks, 3-D printed, and used to shoot silicone parts, a compact way to learn parting lines, draft, alignment and gating on something small enough to iterate in a day.

INJECTION MOLDING

MOLD DESIGN

SOLIDWORKS ASSEMBLIES

SILICONE

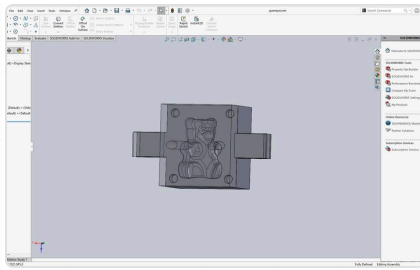
THE EXERCISE

Designing a part and designing the thing that makes the part are different skills. A mold forces you to think in negatives: where does the parting line fall, which way does the material draft, how does the tool stay aligned under pressure, and how does the finished part come out without tearing.

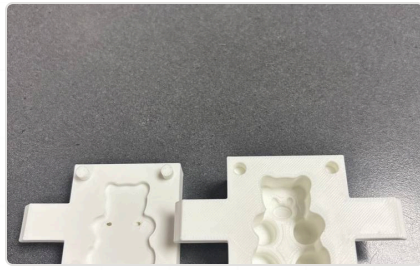
A gummy bear was the ideal test subject: undercuts, compound curves, and no flat surfaces to hide behind.

WHAT IT TOOK

- Modeled the bear, then derived the **cavity and core as a mated SolidWorks assembly** so both halves stayed driven by the same geometry.
- Placed the parting line along the bear's silhouette to keep the undercuts on the arms and ears releasable.
- Added **four corner alignment pins** and matching bosses to keep the halves registered, plus side flanges to clamp against.
- Printed both halves, shot silicone, and demolded, checking flash at the parting line and fill in the ears and feet, the last places to pack out.



In CAD. Cavity and core as one fully defined assembly.



Printed halves. Parting line follows the silhouette; four corner pins keep registration.



Demolded. Ears and feet packed out cleanly, the last features to fill.

Trusting the number before trusting the result

Calibrating eight thermocouple channels against a boiling reference, then using them to map hot spots inside a bare prototype and quantify what adding insulation actually bought.

INSTRUMENTATION

THERMOCOUPLES

DATAQ DI-808

DATA ANALYSIS

THERMAL TESTING

CALIBRATE FIRST

Eight probes reading eight different numbers tells you nothing until you know how much of that difference is the appliance and how much is the probes. So before any hot-spot mapping, all eight channels went into a common boiling reference and logged together at 2 Hz.

The run below is that check: ambient to plateau in about four minutes, held for five, then a slow cool as the bath gave up heat. The useful output isn't the plateau value. It's the **spread** between channels, and the systematic offset from the 212 °F reference.

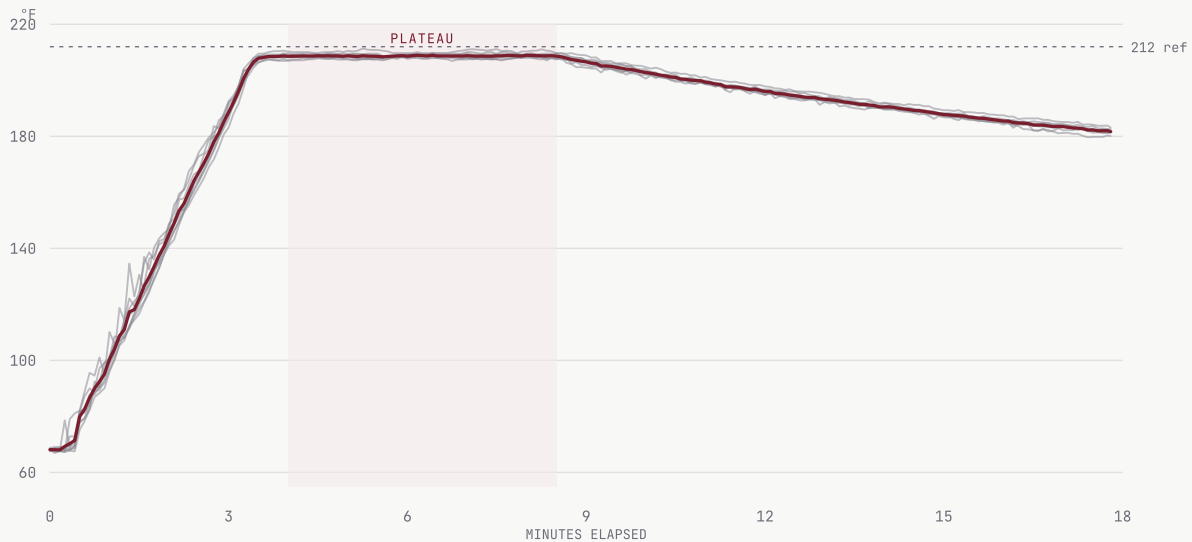
WHAT IT SHOWED

- All eight channels agreed within **2.5 °F** of one another across the plateau, tight enough to compare positions inside a cavity with confidence.
- Each individual channel was stable to under **0.7 °F** of noise, so a real thermal gradient wouldn't be lost in scatter.
- Every channel read low against the 212 °F reference by roughly **3 °F**, a systematic offset, correctable in post rather than a reason to distrust the rig.
- One channel ("Top Lid") sat consistently highest and two sat lowest, so those positions got noted rather than treated as real hot and cold spots.

Eight channels, one boiling reference

Logged at 0.5 s intervals · DATAQ DI-808 · type-K thermocouples. Trace ends where the probes left the bath at 17.8 min.

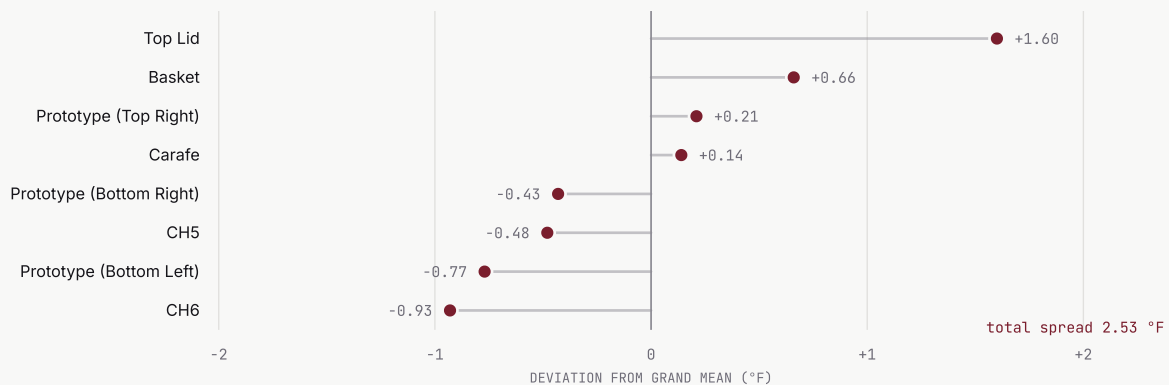
— Individual probes (8) — Mean of all channels



► View plateau values as a table

Channel-to-channel deviation at the plateau

Each channel's mean minus the grand mean, over the 4.0–8.5 minute hold



2.53 °F

Total spread across
all 8 channels at
plateau

<0.7 °F

Per-channel noise,
stable enough to
resolve real
gradients

3.3 °F

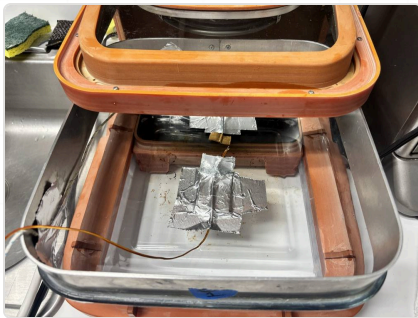
Systematic offset
from the 212 °F
reference

2,137

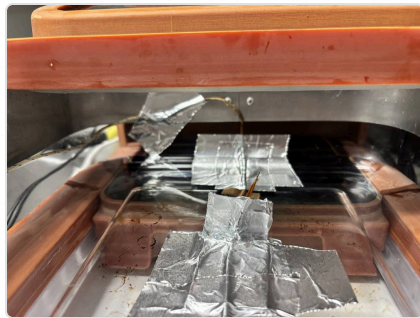
Samples logged per
channel, in bath

THEN: MAPPING THE CAVITY

With the probes trusted, they went into a bare-bones grill and air-fry prototype, foil-taped at fixed positions across the upper and lower cook surfaces so each run measured the same points, and the wire routing was kept identical between runs so it never became a variable.



Probes in position. Foil-taped at fixed points so runs stay comparable.



Both platens at once. Captures the vertical gradient in one run.



Insulation shroud. Built over the bare housing, leads routed identically each run.

AND: WHAT INSULATION BOUGHT

The prototype ran with an uninsulated housing, so heat left through the top as fast as the element put it in. I built a cardboard insulation shroud over the upper cavity, re-ran the same instrumented profile, and compared curves against the bare baseline to quantify the improvement in cook performance.

Letting the food settle the argument

Benchmarking a prototype grill against competitor appliances, and checking whether running its air-fry drawer at the same time cost the grill anything, by cooking the same food the same way under fixed conditions.

COMPETITIVE BENCHMARKING

TEST PROTOCOL

DESIGN RECOMMENDATIONS

THE METHOD

Cook-quality claims fall apart without controls. Every unit got the same cut of pork, the same thickness, the same starting temperature, the same time, and every result was photographed from the same angle under the same light, laid out in the same order.

Once the images are directly comparable, the differences are obvious to anyone in the room, and the conversation moves straight to *why*.

The prototype grills on a top platen and air-fries in a drawer underneath, so it can run both at once. To find out whether that shared load hurt the grill, one condition cooked chops on the grill alone and another cooked the same chops while a batch of tater tots ran in the drawer below.

WHAT CAME OUT OF IT

- Ran the prototype in two conditions, **grill alone** and **grill with the air-fry drawer loaded**, and benchmarked both against two competitor units.
- Scored mark definition, edge-to-edge browning coverage, and uniformity across the four pieces in each batch.
- Found the grill produced essentially the same sear whether or not the drawer was running underneath, so a user can cook both at once without giving anything up on the grill side.
- Fed the findings back as design recommendations alongside the thermal data from project 04, so the visual result and the measured cavity behavior told the same story.



Prototype – Grill only



Prototype – Grill and Air fry



Competitor A



Competitor B

Four units, same protein, same protocol. The prototype's two rows are the grill alone and the grill with tater tots air-frying in the drawer below it, and the browning is effectively unchanged between them.

ALSO

Field work, exhibits, and code.

A co-op in municipal infrastructure and a couple of academic projects that taught me the parts of engineering that aren't CAD: field data, reporting, and working out what a user actually needs.

Boston Water and Sewer Commission

DESIGN ENGINEER CO-OP

- Surveyed 15+ miles of water and sewer infrastructure with a total station, improving municipal spatial data accuracy.
- Integrated infrastructure datasets into the Capital Improvements Program, supporting decisions on \$2M+ of capital investment.
- Gathered the survey data behind 50+ AutoCAD engineering drawings.
- Audited assets in ArcGIS and flagged 25+ record discrepancies.

Boston, MA · January – June 2025

Save the Monkeys

INTERACTIVE MUSEUM EXHIBIT ·
NORTHEASTERN

- Built a working children's museum exhibit teaching permaculture and sustainability.
- Measured exhibit effectiveness with an Excel-based visitor feedback system, a 92% success rate.
- Documented a two-month team project in a 100-page final report.
- Fabricated with SolidWorks, laser cutting and 3-D printing.

Spring 2024

Fourdle

WORDLE-INSPIRED GAME · C++

- Implemented the core game loop and random word generation with switch cases and control structures.
- Built a clear text interface giving players unambiguous feedback on each guess.
- Handled input validation and edge cases as a first real exercise in defensive programming.

Spring 2024

CAD & DESIGN

SolidWorks (parts, assemblies, mold design) · AutoCAD · Adobe Illustrator · ArcGIS

TEST & ANALYSIS

Thermocouple instrumentation · DATAQ data logging · Competitive benchmarking · MATLAB · Excel

PROTOTYPING & SHOP

FDM 3-D printing · Injection molding · Laser cutter · Vinyl cutter · Total station · Soldering · Sewing & embroidery machines

COURSEWORK

Thermodynamics · Fluid Mechanics · Dynamics · Mechanics of Materials · Color & Composition · Design Process & Practice · Representational Drawing

THE OTHER HALF

I sew as much as I CAD.

Curtains, self-drafted patterns, and a lot of thrifted clothing cut apart and rebuilt to fit. Same habit of measuring carefully and iterating until it works, with fabric instead of filament.

[See the sewing work →](#)



CONTACT

Let's build something that has to work.

I'm looking for mechanical engineering and product design co-ops and internships where I can model it, build it, and then go find out how it fails. Robotics especially.

[Email me](#)[LinkedIn](#)

April Norton · Boston, MA

Mechanical Engineering & Design · Northeastern University

I can make *whatever* I think of.

That is the whole reason I sew. If I can picture a thing, I can usually work out how to cut it and put it together, and a week later I am wearing it or it is hanging in my parents' kitchen.

measure twice, seam rip once.



Same instincts, softer materials.

Sewing is where I learned to measure carefully, fit something to a real body or a real window, and accept that the first version is a muslin, not a failure. It is the same loop I run at work with CAD and a 3-D printer, just with fabric and a seam ripper instead of PLA and a deburring tool.

COMMISSION

Curtains for my parents' house

My parents asked, which made this the first sewing project I had to get right the first time. Someone else's windows, someone else's house, no room to shrug and call it a lesson.

WHAT IT TOOK

Curtains look like the easiest thing in the world until you are standing on a chair with a tape measure working out how much fabric a gathered heading actually eats. Every panel had to be measured against its own window, not against the one next to it.

The rest was pinning. A lot of precise, slow pinning, because a hem that drifts a quarter inch over six feet reads as crooked from across the room.

THE DETAILS

- Measured each opening individually and added allowance for the gather, the rod pocket and the hem.
- Pinned and pressed every hem before sewing so nothing wandered under the needle.
- Two different rooms, two different fabrics: an open weave natural linen for the kitchen pass through, a soft floral for the living room.
- Hung and checked in place, because the only tolerance that matters is how it looks on the rod.

MEASURING

HEMMING

MADE TO FIT



Kitchen pass through. Gathered café panels in an open weave, sized so the light still comes through.



Living room panels. Full length, where a quarter inch of hem drift shows from across the room.



A lounge set reverse engineered from my own closet

Two pieces, two completely different approaches to getting a pattern: one traced off a garment I already owned, one printed and taped together from a real pattern for the first time.

TWO WAYS TO GET A PATTERN

For the shorts I took a pair I already own and liked the fit of, traced around them, and added seam allowance. That is reverse engineering: start from the object that already works and recover the geometry that produced it.

The long sleeve top was my **first time actually using a pattern**. I printed it across a stack of pages and taped them into one full size sheet before cutting. Lining up tiled pages is finicky in exactly the way a printed part is finicky, and if the registration is off at the start, nothing downstream fits.

HOW I NORMALLY WORK

Most of the time I do not use a pattern at all. I cut, try it on, pinch where it is wrong, take it in, try it on again. Trial and error on a live fit test, which is slower on paper and much faster in practice.

- Shorts: traced from an existing pair, seam allowance added by hand.
- Top: tiled pattern printed, taped and cut, keeping the registration marks aligned.
- Matched fabric across both pieces so they read as a set.
- Fit adjusted on the body rather than on the pattern, which is still how I prefer to do it.

PATTERN DRAFTING

REVERSE ENGINEERING

KNIT FABRIC



The finished set. Top from a tiled printed pattern, shorts traced off a pair I already owned.



🧩 SOLVING A SMALL ANNOYANCE

A jewelry pouch that stops necklaces tangling

A drawstring pouch with individual slots sewn into the lining, so every necklace gets its own channel and nothing comes out of a bag as one knot.

THE PROBLEM

Necklaces thrown into any soft bag will find each other and tangle. The fix is not a better bag, it is stopping the chains from sharing a volume in the first place.

So the lining is divided into separate slots. Each chain sits in its own channel and the drawstring closes over the top of all of them.

THE BUILD

- Cream floral outer with a pink gingham lining, because the inside should be as nice as the outside.
- Individual sewn slots sized to hold one chain each.
- Drawstring casing at the top so it cinches shut and travels flat.
- Small enough to go in a bag without becoming a project to pack.

LINING CONSTRUCTION

PROBLEM SOLVING



Open, with the slots visible. Gingham lining, one channel per chain.

ONGOING

Most of what I make starts as something else.

A lot of my projects are refashions. I take clothes I already have or something I thrifted and cut it into a new piece, or take it apart and put it back together so it actually fits me. It is the cheapest possible way to

iterate, and it means I am never starting from a blank page or a full price bolt of fabric.

CONTACT

Two halves of the same habit.

If you want the version of this with thermocouples and CAD instead of gingham, the engineering side is one click away.

[See the engineering work](#)

[Email me](#)

April Norton · Boston, MA

Mechanical Engineering & Design · Northeastern University